

Machine Learning Expected Goals Models in Soccer: Reproducible Shot-Outcome Prediction with StatsBomb Data

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Abstract

A fully reproducible expected-goals (xG) modelling pipeline is presented as an interpretable machine learning approach using open football event data from StatsBomb for La Liga 2015/2016 and the 2018 FIFA World Cup. The dataset comprised 10,709 non-penalty shots, with features derived from shot distance and angle, body part (head vs. foot), and competition. Logistic regression and a generalized linear mixed-effects model with shooter-level random intercepts were estimated to predict goal probabilities at the shot level. Model performance was evaluated using information criteria and the area under the ROC curve (AUC). Distance strongly reduced scoring probability, headers were markedly less likely to result in goals than footed shots, and World Cup shots had lower baseline conversion than La Liga attempts at comparable locations. AUC values ranged from 0.75 (baseline location model) to 0.79 (mixed-effects model), providing competitive baselines for soccer prediction tasks. The pipeline illustrates how openly available event data can support transparent, interpretable, and practically useful xG models for research, teaching, and applied coaching and performance analysis in soccer.

Trial registration: not applicable

1 Introduction

Expected-goals (xG) models have become a central tool in football analytics for quantifying the quality of individual shots and evaluating team and player performance over time, building on early work that modelled possessions and shot quality using Markov chains [1–3]. By estimating the probability that a given attempt results in a goal, xG provides a chance-adjusted benchmark that is now routinely used to assess finishing, chance creation, and tactical effectiveness within professional clubs and media coverage [4, 5]. Methodologically, most xG models treat shots as Bernoulli trials and apply generalized linear or machine-learning models with predictors such as shot location, angle, body part, and contextual variables [1, 6]. Recent summaries of xG modelling highlight a progression from simple location-based specifications to models that incorporate player ability, contextual factors, and temporal information about the build-up to shots [7, 8].

Recent work has advanced xG modelling in several directions. Bayesian hierarchical approaches such as Bayes-xG incorporate player and positional random-effects to separate chance from persistent scoring ability [9]. Other studies propose interpretable generalized additive or Bayesian mixed-effects models that retain competitive predictive performance while allowing direct examination of distance, angle, and player-level effects [9, 10]. Together, these developments highlight both the predictive value of xG and the importance of modelling heterogeneity across players and teams.

However, much of this literature relies on proprietary event and tracking data from commercial providers, which cannot be freely shared with researchers, students, or practitioners outside contracted organisations [11, 12]. This dependence on closed data limits transparency, replication, and pedagogy, echoing broader calls for reproducible computational workflows in data-intensive science [13]. Although some recent work has demonstrated that event-level structures can be inferred directly from tracking data using open-source implementations [14], and vendor-hosted open resources such as StatsBomb Open Data and the Sportec Open DFL tracking dataset have begun to appear, these sources typically require substantial preprocessing and careful documentation to support reproducible research [15, 16]. A growing ecosystem of free football data sources now exists, including curated lists of open event and tracking datasets, but there remains a need for worked examples that show how to construct publishable, transparent xG models from these files and apply them in coaching and performance-analysis contexts.

The present study addresses this need by developing a fully reproducible xG pipeline based solely on public event data from StatsBomb Open Data, focusing on La Liga 2015/2016 and the 2018 FIFA World Cup [17]. Specifically, the analysis (a) constructs shot-level features from event files, including distance and angle to goal, body part, and competition indicators; (b) estimates fixed-effects logistic regression models and shooter-level mixed-effects models; and (c) evaluates model fit and discrimination using deviance, information criteria, and receiver operating characteristic analysis, while explicitly quantifying what can be achieved with thousands of shots from openly available competitions. The aim is not to introduce a new state-of-the-art xG algorithm, but rather to provide an open, interpretable, and practically usable template that coaches, performance analysts, and researchers can adopt and extend for chance-adjusted evaluation in elite football.

2 Methods

2.1 Data source and context

We analysed open football event data from multiple professional competitions provided by StatsBomb through their public *StatsBomb Open Data* repository on GitHub [18]. This repository includes standardized JSON event files for a range of domestic leagues and international tournaments, accompanied by documentation that specifies event types, locations, and contextual qualifiers such as body part and shot outcome. For the present study, we focused on men's elite competitions with complete event coverage and selected La Liga 2015/2016 and the 2018 FIFA World Cup as two high-profile contexts that combine tournament and league play and are explicitly listed among the open competitions in the repository. Event data for all matches in these competitions were downloaded from the

repository and merged into a shot-level dataset using competition, season, match, and player identifiers, following approaches adopted in existing open xG tutorials that work directly with StatsBomb data [8, 9].

2.2 Sample and inclusion criteria

From the StatsBomb event files for La Liga 2015/2016 and the 2018 FIFA World Cup, we extracted all actions coded as shots, defined as events with `type.name = "Shot"` in the StatsBomb schema. To focus on non-penalty scoring chances, we excluded penalties, identified as events with `shot.type.name = "Penalty"`, and removed own goals, yielding a dataset of non-penalty shots with clearly defined shooters, locations, and outcomes. For each qualifying shot, we retained the team in possession, competition and season identifiers, match identifier, period, game minute, shooter and goalkeeper identifiers, the two-dimensional shot location on the pitch, shot outcome (goal vs. non-goal), body part, and a small set of contextual qualifiers (e.g., open play vs. set piece).

To facilitate modelling, we oriented the pitch so that all shots were treated as attacking from left to right toward the same goal, mapping StatsBomb's original 120×80 coordinate system into a consistent reference frame across matches and competitions. This ensured a single fixed goal location for the computation of distance and angle variables and is consistent with prior work that constructs xG models from StatsBomb locations [9, 19].

2.3 Measures and variable construction

All data processing and feature engineering were conducted in R (Version 4.5.2) using the tidyverse suite for data manipulation and visualisation [20]. Access to and parsing of the StatsBomb JSON files followed published guidance for working with StatsBomb data in R, including extracting event-level information and reshaping it to the shot level. Shot locations were obtained from the StatsBomb location field, which records the (x,y) coordinates of the ball at the time of the shot on a 120×80 pitch; these coordinates were kept on their original scale for distance and angle calculations and rescaled to the unit interval only for visualisation. Following the StatsBomb specification, we treated the centre of the attacking goal as (120, 40) in the original coordinate system and applied the same convention after re-orientation.

For each shot i , we defined the following variables.

Shot distance to goal. We computed the Euclidean distance from the shot location to the goal centre, a standard predictor in xG models that is strongly associated with scoring probability [12, 21].

Shot angle. We calculated the angle between the line segment from the shot location to the goal centre and the horizontal axis, expressed in radians, such that shots taken from more central positions have angles closer to 0 and typically correspond to higher scoring chances [21, 22].

Outcome. The dependent variable was a binary indicator coded as 1 when the StatsBomb `shot.outcome.name` field indicated "Goal" and 0 for all other non-penalty outcomes (on-target saves, off-target, and blocked shots).

Body part. We derived a categorical variable indicating the body part used to take the shot from `shot.body_part.name`, contrasting headers with all footed shots, consistent with prior evidence that scoring probabilities differ systematically by body part [23, 24].

Competition and match indicators. Categorical factors indicated the competition (World Cup vs. La Liga), season, and match, allowing us to test for between-competition differences in baseline scoring probabilities and, when needed, to account for within-match clustering [25, 26].

We did not standardise predictors because the pitch coordinates are inherently bounded and the raw distance and angle metrics are directly interpretable on the native spatial scale. All transformation and feature-engineering steps were documented in an openly available R script to support exact computational reproducibility.

2.4 Statistical modelling strategy

2.4.1 Fixed-effects logistic regression models

Let Y_i denote the binary outcome for shot i (1 = goal, 0 = non-goal) and π_i the corresponding goal probability. In line with the Machine Learning in Soccer agenda, we treat expected goals models as interpretable machine learning classifiers for shot-level goal prediction in soccer. Specifically, we use logistic regression and generalized linear mixed-effects models as baseline machine learning approaches that map shot features (distance, angle, body part, competition) to goal probabilities. We first estimated a baseline fixed-effects logistic regression including only distance and angle:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{shot_dist}_i + \beta_2 \text{shot_angle}_i.$$

This specification mirrors baseline xG models that rely primarily on spatial shot characteristics [9, 19]. We then extended the model to include body part and competition indicators as additional fixed-effects:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{shot_dist}_i + \beta_2 \text{shot_angle}_i + \beta_3 \text{body_part}_i + \beta_4 \text{competition}_i.$$

In this model, footed shots and La Liga matches served as reference categories. Including body part and competition is consistent with xG models that add contextual covariates to improve interpretability and, where sample size permits, predictive performance [12, 19]. All fixed-effects models were estimated in R using the `glm` function with binomial family and logit link.

2.4.2 Mixed-effects model with shooter random intercepts

To explore potential between-shooter variability in scoring probabilities, we specified a generalized linear mixed-effects model (GLMM) with random intercepts for shooter identity, paralleling hierarchical xG models that include player-level random-effects to separate chance from persistent skill differences [9, 27]. Let $j[i]$ index the shooter who took shot i . The GLMM is

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{shot_dist}_i + \beta_2 \text{shot_angle}_i + \beta_3 \text{body_part}_i + \beta_4 \text{competition}_i + u_{j[i]}$$

where u_j is a random intercept for shooter j , with $u_j \sim N(0, \sigma_u^2)$. Shooter identity was derived from the StatsBomb `player.id` field and treated as a factor.

2.4.3 Model evaluation and diagnostics

We evaluated and compared models using parameter inspection, likelihood-based fit indices, and discrimination metrics, following common practice in xG and generalized linear modelling [8, 12]. For each model, we examined estimated coefficients, standard errors, Wald z statistics, and associated p values. We compared nested fixed-effects models (baseline vs. extended) using changes in deviance and Akaike’s Information Criterion (AIC), and we also reported the Bayesian Information Criterion (BIC) to account for model complexity [28]. For the mixed-effects model, we inspected the estimated variance of the shooter random intercept and checked for boundary (singular) fits to assess whether the data supported a non-zero random-effects component.

To quantify predictive discrimination, we computed the area under the receiver operating characteristic curve (AUC) for each model using the `pROC` package in R [29]. AUC is widely used to evaluate probabilistic classifiers, including xG models that output goal probabilities instead of hard labels [8, 19]. For descriptive interpretation, we also calculated mean predicted xG separately for goals and non-goals and produced partial effect plots of shot distance versus predicted scoring probability, holding other predictors constant [21]. Throughout, we emphasised effect sizes, uncertainty, and model behaviour over dichotomous significance testing and aligned our workflow with general recommendations for reproducible computational research [30, 31].

3 Results

3.1 Descriptive statistics

Across La Liga 2015/2016 and the 2018 FIFA World Cup in the public StatsBomb Open Data, the final dataset contained 10,709 non-penalty shots, of which 1,080 resulted in goals (overall conversion rate = 10.1%;). Conversion rates were slightly higher in La Liga (945 goals from 9,071 shots; 10.4%) than in the World Cup (135 goals from 1,638 shots; 8.2%). Mean shot distance was approximately 19.1 StatsBomb pitch units in La Liga and 19.3 in the World Cup (both SD = 8.7–8.8), and mean shot angles were close to zero radians in both competitions (La Liga: M = 0.02, SD = 0.60; World Cup: M = −0.01, SD = 0.61), indicating broadly similar spatial distributions of attempts across contexts.

Table 1 summarises the distribution of non-penalty shots by competition, body part, and outcome.

<<insert Table 1 here>>

Table 1 <i>Shot characteristics by competition</i>							
Competition	Shots (n)	Goals (n)	Conversion rate	Mean distance	SD distance	Mean angle	SD angle
La Liga 2015/2016	9,071	945	0.104	19.10	8.70	0.02	0.60
World Cup 2018	1,638	135	0.082	19.30	8.80	−0.01	0.61
<i>Note.</i> Values are based on non-penalty shots extracted from StatsBomb Open Data after re-orienting all pitches so that teams attack from left to right and treating the centre of the attacking goal as (120, 40) in the native 120 × 80 coordinate system.							

A scatterplot of realised outcomes against shot distance, overlaid with the fitted logistic curve from the baseline model, showed that goals tended to occur from shorter distances, although there remained substantial overlap between goals and non-goals at intermediate ranges, consistent with prior xG work using large samples [19, 32].

Figure 1 illustrates the spatial distribution of shot locations for La Liga and World Cup matches on the standardized pitch.

<<insert Fig. 1 here>>

3.2 Logistic models: coefficients

In the baseline logistic regression including only shot distance and angle, the intercept was not distinguishable from zero, but distance showed a clear negative association with scoring probability, whereas angle was not associated with outcomes once distance was controlled. The intercept was $\beta_0 = -0.07$ (SE = 0.08, $p = .40$), shot distance $\beta_1 = -0.13$ (SE = 0.01, $p < .001$), and shot angle $\beta_2 = -0.02$ (SE = 0.05, $p = .69$), with a residual deviance of 6,161.8 on 10,706 degrees of freedom (AIC = 6,167.8). These patterns align with previous findings that distance is a dominant determinant of xG, whereas simple angle terms may add limited information once distance is accounted for [9, 12].

Extending the model to include body part and competition yielded more precise estimates and clear effects for headers and competition context. In this extended fixed-effects model, the intercept was $\beta_0 = 0.71$ (SE = 0.10, $p < .001$), shot distance $\beta_1 = -0.17$ (SE = 0.01, $p < .001$), shot angle $\beta_2 = -0.05$ (SE = 0.05, $p = .38$), body part (head vs. foot) $\beta_3 = -1.16$ (SE = 0.09, $p < .001$), and competition (World Cup vs. La Liga) $\beta_4 = -0.25$ (SE = 0.10, $p = .02$). The null deviance was 7,002.5 on 10,708 degrees of freedom and the residual deviance was 5,970.0 on 10,704 degrees of freedom (AIC = 5,980.0). Exponentiating the body-part coefficient implies that, holding distance and angle constant, headers have odds of scoring that are roughly 31% of those for footed shots, which is in line with prior evidence that headers are generally harder to convert than shots with the feet [24, 33].

As shown in Table 2, shot distance was a strong negative predictor of scoring probability in both models, whereas shot angle was not statistically significant.

<<insert Table 2 here>>

Table 2 Logistic regression coefficients for baseline and extended models				
Model / Predictor	Estimate	SE	z	p
Baseline (distance + angle)				
Intercept	−0.07	0.08	−0.84	.404
Shot distance	−0.13	0.01	−25.01	< .001
Shot angle	−0.02	0.05	−0.40	.688
Extended (+ body part, comp.)				
Intercept	0.71	0.10	7.38	< .001
Shot distance	−0.17	0.01	−27.59	< .001
Shot angle	−0.05	0.05	−0.89	.375
Body part (head vs. foot)	−1.16	0.09	−12.78	< .001
Competition (World Cup vs. La Liga)	−0.25	0.10	−2.42	.015
<i>Note.</i> Outcome is goal (1) vs. non-goal (0). Predictors are distance and angle to goal, body part (head vs. foot), and competition (World Cup vs. La Liga), computed from StatsBomb locations as described in the Method section.				

Mean predicted xG values were higher for goals than for non-goals in all specifications. For the baseline model, mean predicted xG was approximately 0.18 for goals versus 0.09 for non-goals, while in the extended fixed-effects model the corresponding means were 0.20 and 0.09, respectively. In the mixed-effects model (see below), these means increased further to 0.21 for goals and 0.09 for non-goals. This pattern indicates that all models assigned larger probabilities, on average, to shots that resulted in goals, as expected for xG models built on substantial samples [8].

Figure 2 shows mean predicted expected-goals (xG) values for shots that did not result in goals and shots that resulted in goals, for the baseline, extended fixed-effects, and mixed-effects models.

<<insert Fig. 2 here>>

3.3 Mixed-effects model and model comparison

The generalized linear mixed-effects model with shooter-level random intercepts and the same fixed-effects produced similar fixed-effects estimates but a non-zero random-effects variance for players. The shooter random-intercept variance was approximately 0.10 (SD = 0.31), indicating meaningful between-player differences in baseline scoring tendencies after adjusting for distance, angle, body part, and competition. Fixed-effects estimates remained very close to those in the extended logistic model, with shot distance and body part retaining strong effects and the competition effect becoming marginal.

Model comparison using information criteria favoured richer specifications over the simplest baseline. AIC values were 6,167.8 for the baseline model, 5,980.0 for the extended fixed-effects model, and 5,967.7 for the mixed-effects model, indicating a substantial improvement when adding body part and competition and a smaller additional improvement when allowing shooter-level random intercepts. BIC values preferred the extended fixed-effects model over the baseline, reflecting the better fit despite additional

parameters. These patterns are consistent with recommendations to select xG models based on penalised fit indices rather than raw deviance alone when multiple plausible structures are available [34, 35].

Table 3 summarises model comparison statistics (degrees of freedom, AIC, and BIC) for the baseline, extended fixed-effects, and mixed-effects models.

<<insert Table 3 here>>

Table 3 Model comparison by AIC and BIC			
Model	df	AIC	BIC
Baseline (dist + angle)	3	6,167.8	6,189.6
Fixed (+ body part, comp.)	5	5,980.0	6,016.4
Mixed (+ shooter random)	6	5,967.7	—
Note. Lower AIC/BIC values indicate a better trade-off between fit and complexity. BIC is reported for the fixed-effects models, while AIC is used as the primary criterion for the mixed-effects model, in line with common practice for GLMMs [36].			

3.4 Discrimination: ROC curves and AUC

In the full 10,709-shot sample, the baseline distance-and-angle model achieved an area under the ROC curve (AUC) of 0.75, indicating moderate discrimination between goals and non-goals [29]. Adding body part and competition improved AUC to 0.78, and including shooter random intercepts yielded a further increase to 0.79. These values are comparable to or slightly below those reported for more complex xG models that incorporate richer contextual features, but they demonstrate that even relatively simple specifications can achieve useful discrimination when estimated on thousands of shots [2, 8].

Figure 3 shows the receiver operating characteristic (ROC) curves for the baseline, extended fixed-effects, and mixed-effects models, with AUC values indicated in the legend.

<<insert Fig. 3 here>>

Table 4 reports the area under the ROC curve (AUC) for the baseline, extended fixed-effects, and mixed-effects models.

<<insert Table 4 >>

Table 4 AUC for fixed- and mixed-effects models	
Model	AUC
Baseline (dist + angle)	0.75
Fixed (+ body part, comp.)	0.78
Mixed (+ shooter random)	0.79
Note. AUC values are based on ROC curves computed using the pROC package in R [29]. Higher values indicate better discrimination between goals and non-goals.	

4 Discussion

The primary goal of this study was to demonstrate that a technically sound xG pipeline can be built entirely from public, elite-level event data while remaining transparent, reproducible, and practically usable. Using StatsBomb Open Data for La Liga 2015/2016 and the 2018 FIFA World Cup, we derived standard spatial features (distance and angle to goal) and simple contextual variables (body part, competition indicators) and fitted logistic and mixed-effects models to predict goal probability. Distance emerged as a strong negative predictor of scoring, headers were substantially less likely to result in goals than footed shots at comparable locations, and shots in World Cup matches had lower baseline conversion than those in La Liga, aligning with prior work that identifies distance, body part, and competition context as key xG determinants [6, 12]. At the same time, simple angle terms contributed little additional information beyond distance, consistent with studies showing that angle effects are often absorbed or refined by richer feature sets [12, 24].

The mixed-effects analyses provide insight into when hierarchical xG models are viable in open-data settings. In contrast to earlier small-sample demonstrations where shooter random-effects variances collapsed to zero and mixed-effects fits were singular, the present dataset supported a non-zero shooter-level variance component, indicating genuine between-player differences in baseline scoring tendencies after adjusting for distance, angle, body part, and competition [9, 27]. Information criteria showed that adding body part and competition markedly improved fit over a location-only specification, and that including shooter random intercepts yielded a smaller but still favourable reduction in AIC, in line with recommendations to align hierarchical model complexity with the volume and granularity of available data [37, 38]. The resulting AUC values in the 0.75–0.79 range are comparable to those reported for relatively

simple xG models and underscore that interpretable structure can achieve useful discrimination without resorting to highly opaque machinery [8, 39].

From an applied perspective, these findings highlight both the value and the constraints of current open football data for performance analysis and coaching support. Public xG studies using vendor-released event datasets and technical-report frameworks have shown that extended feature sets incorporating, for example, preceding event sequences, defensive pressure, and player quality proxies can further improve predictive performance and practical usefulness [2, 7]. StatsBomb Open Data provide consistent event coding across multiple competitions and seasons, with detailed shot qualifiers and locations that are well suited to reproducible, large-sample modelling. However, the absence of full tracking data in this open repository limits the direct inclusion of spatial defensive pressure, movement-based build-up features, and certain goalkeeper positioning variables that appear in recent hierarchical and Bayesian xG frameworks [40, 41]. Our results therefore reflect what can be achieved with a deliberately parsimonious feature set that is accessible to any researcher, student, coach, or performance analyst willing to work with publicly available JSON files.

A key contribution of this study is conceptual and infrastructural rather than purely predictive. By providing a fully documented pipeline from raw public events to shot-level features, logistic and mixed-effects models, ROC-based evaluation, and a public code repository, we offer a template that can be reused, audited, and scaled as more open data become available [42, 43]. Such pipelines are increasingly important for teaching, for developing analysts within clubs and academies, and for fostering an open, cumulative science of football analytics in which methods and results are not confined to organisations with proprietary feeds. They also complement recent work on interpretable Bayesian xG models, which emphasise understanding model structure and uncertainty alongside headline performance metrics [8, 44]. In this sense, the present analysis illustrates how open event data and openly shared code can bring some of the benefits of advanced xG modelling to broader academic and practitioner communities.

Beyond their value for football analytics, the models developed here provide baseline machine learning classifiers for shot-level goal prediction using only publicly available event data. Aggregated xG outputs from these models can be used as features in downstream match-outcome or tournament-prediction systems, aligning with current work on soccer prediction challenges and ML-based performance analysis. In this sense, our pipeline offers a transparent, reproducible benchmark that the machine learning community can extend with richer architectures, additional features, or alternative loss functions while retaining interpretability.

Several limitations should be acknowledged. First, although our sample (over 10,000 non-penalty shots) is large by open-data standards, it is modest relative to the multi-season, multi-league datasets often used in commercial or proprietary xG work, which can span hundreds of thousands of shots [19, 45]. As a result, we focused on relatively simple fixed-effects structures and only a single level of shooter random-effects, rather than more elaborate hierarchical or interaction-rich models. Second, the feature set was intentionally restrained to distance, angle, body part, and competition; richer contextual variables such as defensive pressure, dynamic build-up information, or psychological factors were not included, even though these have been shown to affect scoring probabilities when appropriate data are available [46, 47]. Third, we evaluated model discrimination using in-sample AUC, because formal out-of-sample validation would require additional design choices (e.g., competition- or season-wise splits) that were beyond the scope of this methodological demonstration but should be considered in future work.

Future research can build directly on this pipeline in several ways. One priority is to aggregate additional open competitions and seasons to approach the scale at which richer hierarchical structures such as player-by-position random slopes or team-level random-effects can be estimated robustly [48, 49]. Another is to integrate contextual features derived from complementary open sources, such as tracking-based estimates of defensive pressure or ball-carrier movement, building on recent work that links event sequences and spatiotemporal information to improved xG predictions [7, 19]. As open data ecosystems mature, there is also scope for community-developed standards that simplify linking events, tracking, and derived tactical metrics, making it easier to share not only code but also complete, reproducible analytical workflows [50, 51]. In this broader context, our contribution is to lower the barrier to entry by showing how a principled xG model can be implemented with nothing more than widely available open event data and standard R packages, and by illustrating how such a model can support chance-adjusted evaluation for coaches and performance analysts in elite football.

4.1 Conclusion

This study showed that it is feasible to build a transparent xG pipeline entirely from publicly available elite-match event data. Using StatsBomb Open Data for La Liga 2015/2016 and the 2018 FIFA World Cup, we derived standard shot-level features and estimated logistic and shooter-level mixed-effects models that produced plausible goal probabilities, non-trivial discrimination between goals and non-goals, and interpretable effects of distance, body part, and competition [8, 9]. In doing so, we provide a worked example of how open event data can support reproducible xG modelling for research, teaching, and applied performance analysis, complementing recent work that has demonstrated the practical value of xG in larger, often proprietary datasets [1, 52]. Taken together, these results position our reproducible xG pipeline as an interpretable machine learning baseline for soccer prediction and performance analysis using open data.

Declarations

Funding

No funding was received to assist with the preparation of this manuscript.

Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Informed consent

The present study relied exclusively on fully de-identified, publicly available data and therefore did not require additional participant consent.

Ethics approval

This study involved secondary analysis of fully anonymised, publicly available football event data obtained from the StatsBomb Open Data repository (<https://github.com/statsbomb/open-data>), which is released under an open data license (CC BY 4.0) for research and educational use. No human participants were directly recruited, and no identifiable personal information was collected or processed at any stage of the analysis. All procedures involving human-derived data were conducted in accordance with the principles of the 1964 Declaration of Helsinki and its subsequent amendments. In line with institutional and international guidelines on the use of secondary, de-identified data, formal ethical approval was not required for the present study.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Author Contribution

K.N.A. conceived and designed the secondary analysis, implemented the ex-Gaussian modeling, conducted the statistical analyses, and drafted the main manuscript text. N.A. contributed to the study design, literature review, and interpretation of the findings. D.K.S. contributed to data preparation, coding decisions, and critical revision of the methods and results. E.E.N. contributed to the conceptual framing, sport-specific interpretation, and substantial editing of the manuscript. All authors reviewed and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

Data Availability

The event data analysed in this study are publicly available from the StatsBomb Open Data repository on GitHub <https://github.com/statsbomb/open-data>, including La Liga 2015/2016 and the 2018 FIFA World Cup. All R scripts used for data processing and analysis are available at <https://github.com/nyantakyiappiah-eng/open-xg-statsbomb-pipeline>, enabling full computational reproducibility.

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Figures

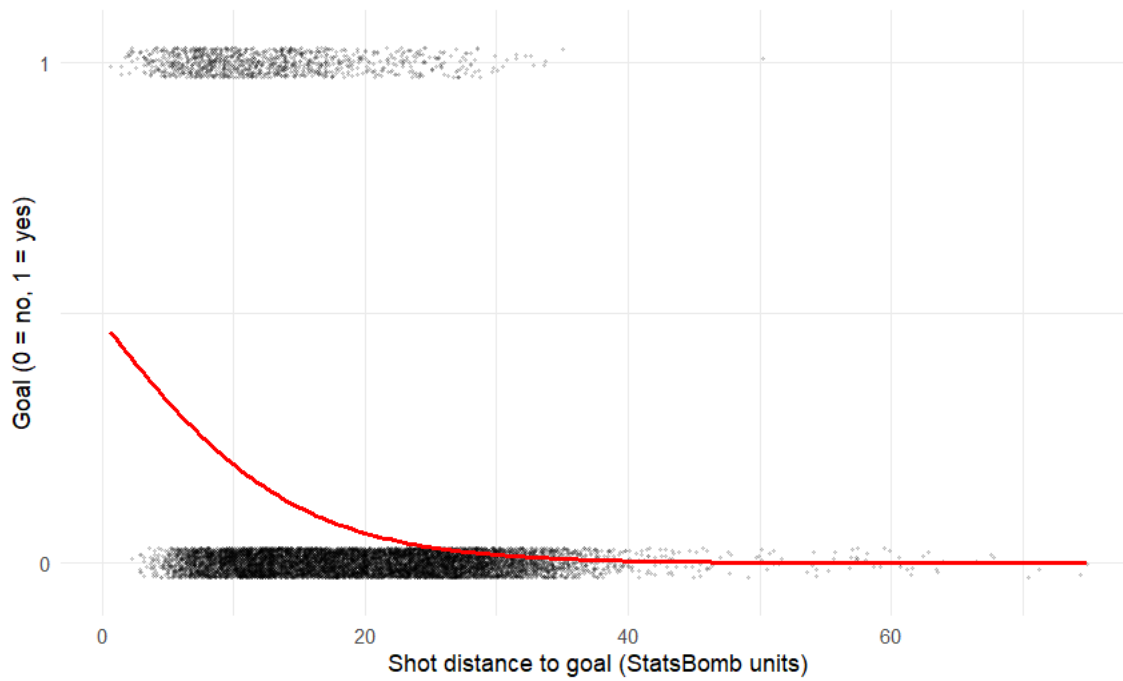


Figure 1

Shot distance and goal outcome for 10,709 non-penalty shots from La Liga 2015/2016 and the 2018 FIFA World Cup, with a fitted logistic curve from the baseline model.

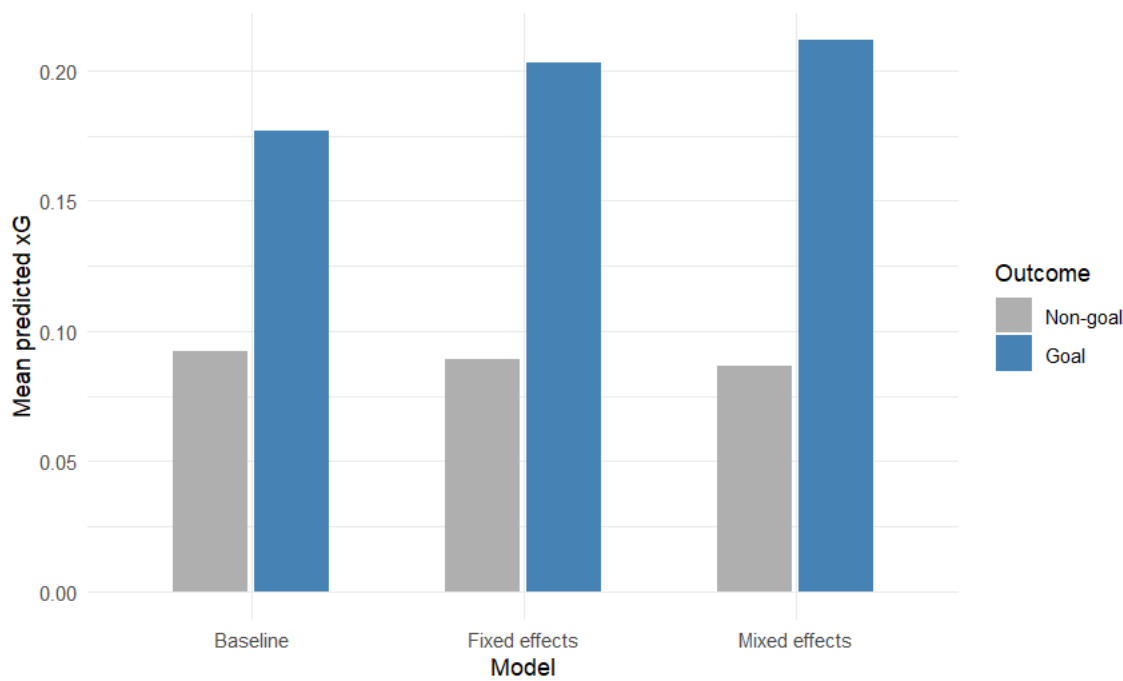


Figure 2

Mean predicted expected-goals (xG) values for shots that did not result in goals and shots that resulted in goals, for the baseline, extended fixed-effects, and mixed-effects models.

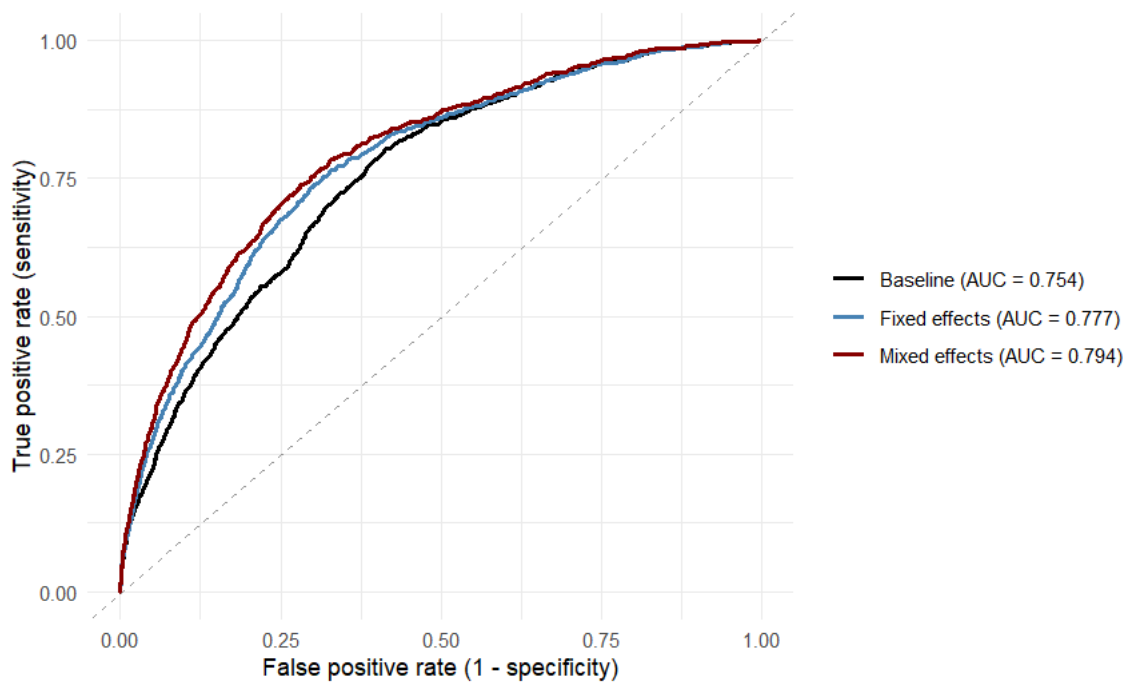


Figure 3

Receiver operating characteristic (ROC) curves for the baseline, extended fixed-effects, and mixed-effects models fitted to 10,709 non-penalty shots, with AUC values indicated in the legend